

Unmanned Aircraft System Mapping of the LNVA Canal Site

Project Summary | Summer 2026

Prepared by

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Site: LNVA canal near Fannett Road, Beaumont, Texas

Project designation: CAWAQ_LNVA

Objective and Scope

Objective

Produce a georeferenced 3D surface model of the LNVA canal site from UAS imagery.

Out of scope

Ground control survey, absolute vertical accuracy validation, and vegetation or structure modelling.

In scope

- Lab and equipment familiarization
- ArcGIS Pro 3.6 training
- Field data collection at the canal site
- Photogrammetric processing in Pix4Dmapper
- Scene correction, OBJ export and publication

Summary of Work Completed

The full scope of summer activity, in sequence

- 1** Completed equipment familiarization in the Geospatial Lab and drone lab, covering battery charging procedure and landing pad operation.
- 2** Completed ArcGIS Pro 3.6 training courses delivered through the Esri learning platform.
- 3** Conducted two test flights at the LNVA canal site and recovered 182 usable images totalling 9.1 GB from the second flight.
- 4** Processed the imagery in Pix4Dmapper to produce a digital surface model, an orthomosaic and a dense point cloud at 0.59 cm ground sample distance.
- 5** Built a local 3D scene in ArcGIS Pro 3.6 with the digital surface model as ground elevation source and the orthomosaic draped over it.
- 6** Applied hydro-flattening to correct the canal water surface within the elevation model.
- 7** Exported the corrected surface to 3D OBJ format through a 7-step geoprocessing chain.
- 8** Published the scene to ArcGIS Online as an optional supporting deliverable.

Equipment and Software Used

Platform and sensor

- Sony ILX-LR1, 61 MP full frame camera
- Flight batteries and field charging
- Portable landing pad

Working coordinate system

WGS 84 / UTM zone 15N throughout.

Processing environment

- Geospatial Lab workstation
- Pix4Dmapper, full 3D Maps template
- ArcGIS Pro 3.6
- ArcGIS Online, Lamar organizational account
- Open3D Viewer

Site and Mission Planning

Mission parameters

- **Site:** LNVA canal, Fannett Road
- **Project:** CAWAQ_LNVA
- **Area:** approximately 8 acres
- **Altitude:** 35 m above ground level
- **Ground speed:** 3.16 m/s
- **Ground sample distance:** 0.59 cm
- **Pattern:** overlapping nadir imagery
- **Coordinate system:** WGS 84 / UTM zone 15N
- **Ground control:** none surveyed



Field Testing Day

LNVA canal site, July 2026

First flight

A 30-minute test flight completed the planned path, with supplementary video recorded. The overlapping image set could not be read due to a format mismatch and was not recoverable.

Second flight

The mission was reflighted for 15 minutes at 35 m above ground level, producing 182 overlapping images totalling 9.1 GB. Readability was verified before leaving the site.

Operational lesson

Confirm image format and readability on site immediately after landing. This converts a lost flight day into a 5-minute check.



Photogrammetric Processing in Pix4Dmapper

Pipeline

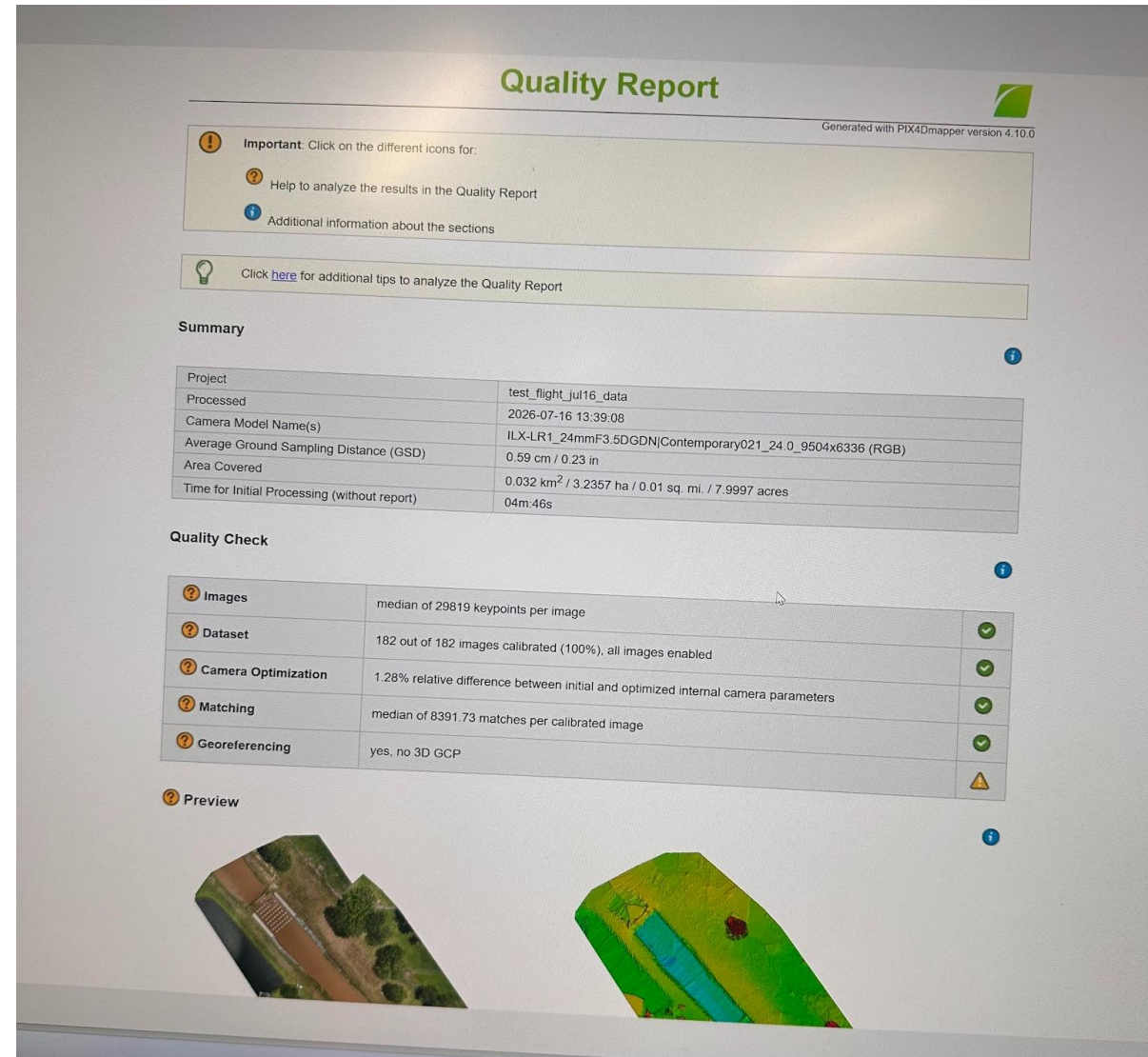
- Initial processing: keypoints and calibration
- Densification: point cloud and mesh
- Digital surface model and orthomosaic

Quality report indicators

- Mean reprojection error of 0.131 pixels
- Geolocation consistent with RTK or PPK grade
- Rolling shutter 26 cm per frame at 3.16 m/s

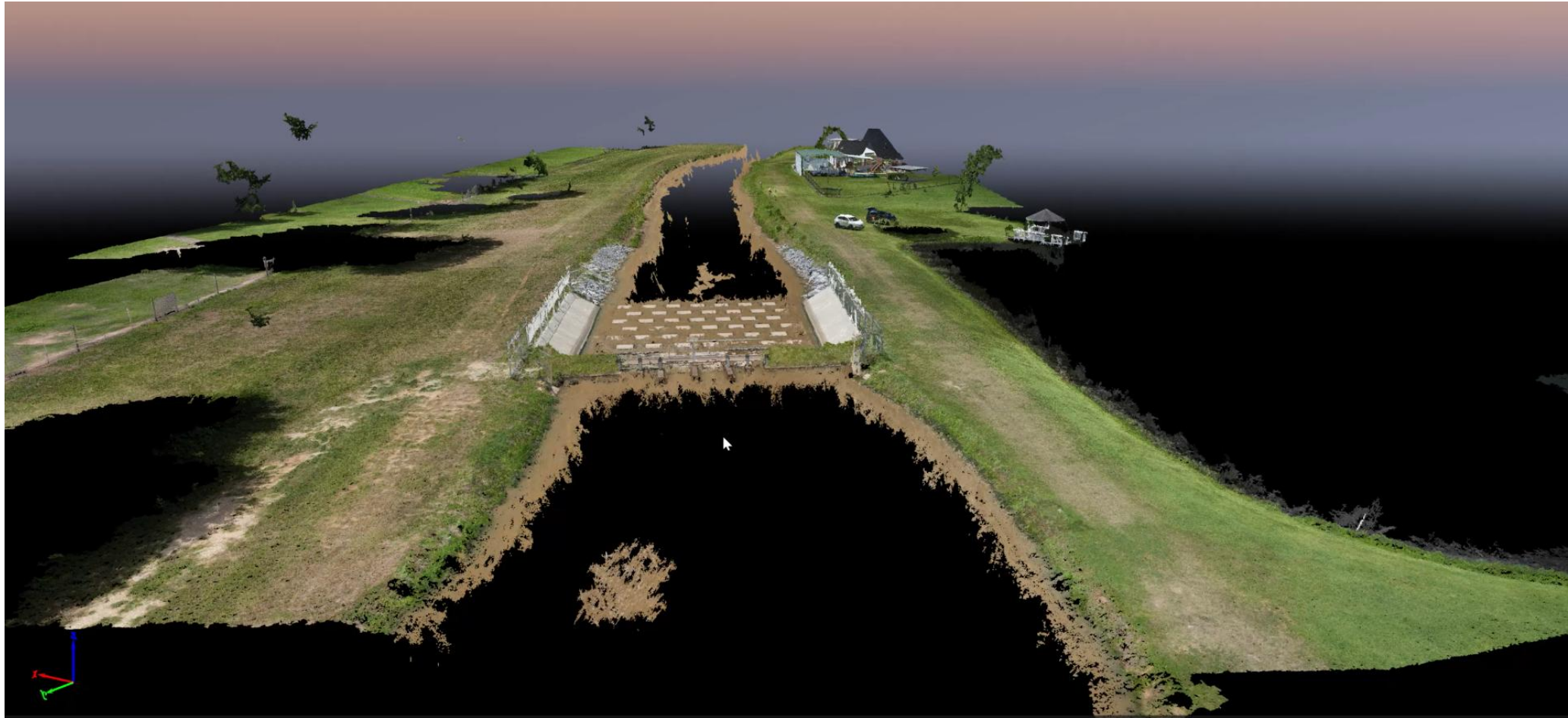
Outputs

Digital surface model and orthomosaic as GeoTIFF,
dense point cloud as LAS.



Processing Result: Surface Model and Orthomosaic

Pix4Dmapper output, WGS 84 / UTM zone 15N, 0.59 cm ground sample distance



Scene Construction in ArcGIS Pro 3.6

Geospatial Lab workstation

Steps taken

- Local scene created in WGS 84 / UTM zone 15N
- Surface model set as ground elevation source
- WorldElevation3D basemap surface removed
- Orthomosaic added and draped on the surface

Why the basemap surface was removed

Leaving the world elevation surface active would drape the orthomosaic over Esri terrain rather than the flight derived surface, producing a plausible but incorrect scene.



Hydro-Flattening of the Canal Surface

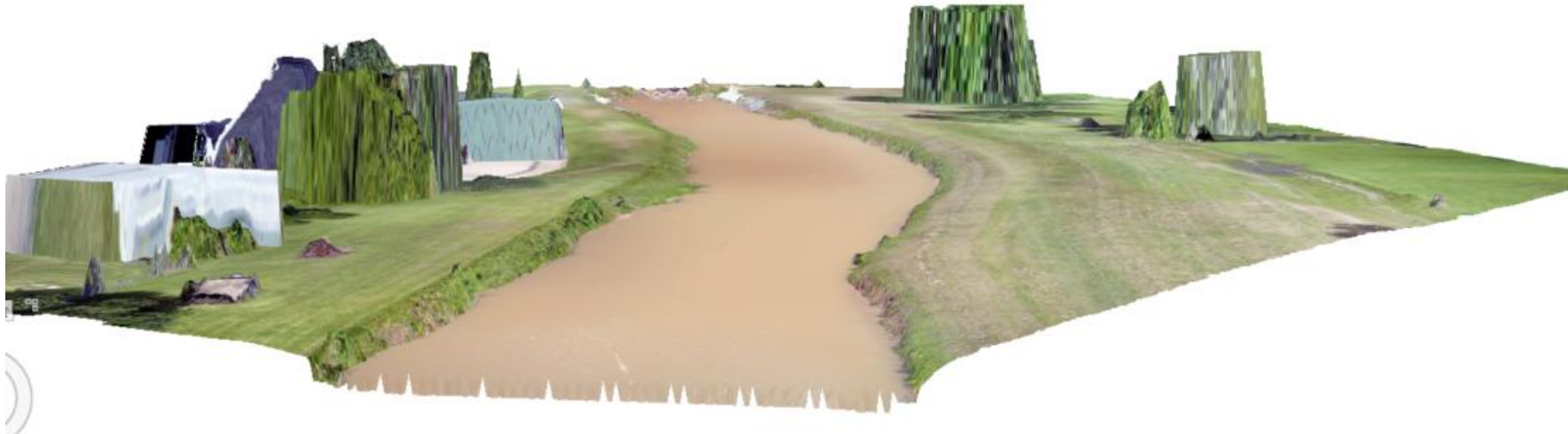
ArcGIS Pro 3.6, 24 to 29 July 2026

Method

- Water polygon digitized with an interior hole preserving the dock
- Zonal Statistics as Table gave a median water elevation near -22.5 m
- Polygon to Raster produced a constant water surface
- Raster Calculator substituted that surface into the model
- Feature Preserving Smoothing reduced residual noise

Iteration

Eight tool runs failed across 24 and 29 July before the conditional expression resolved correctly.

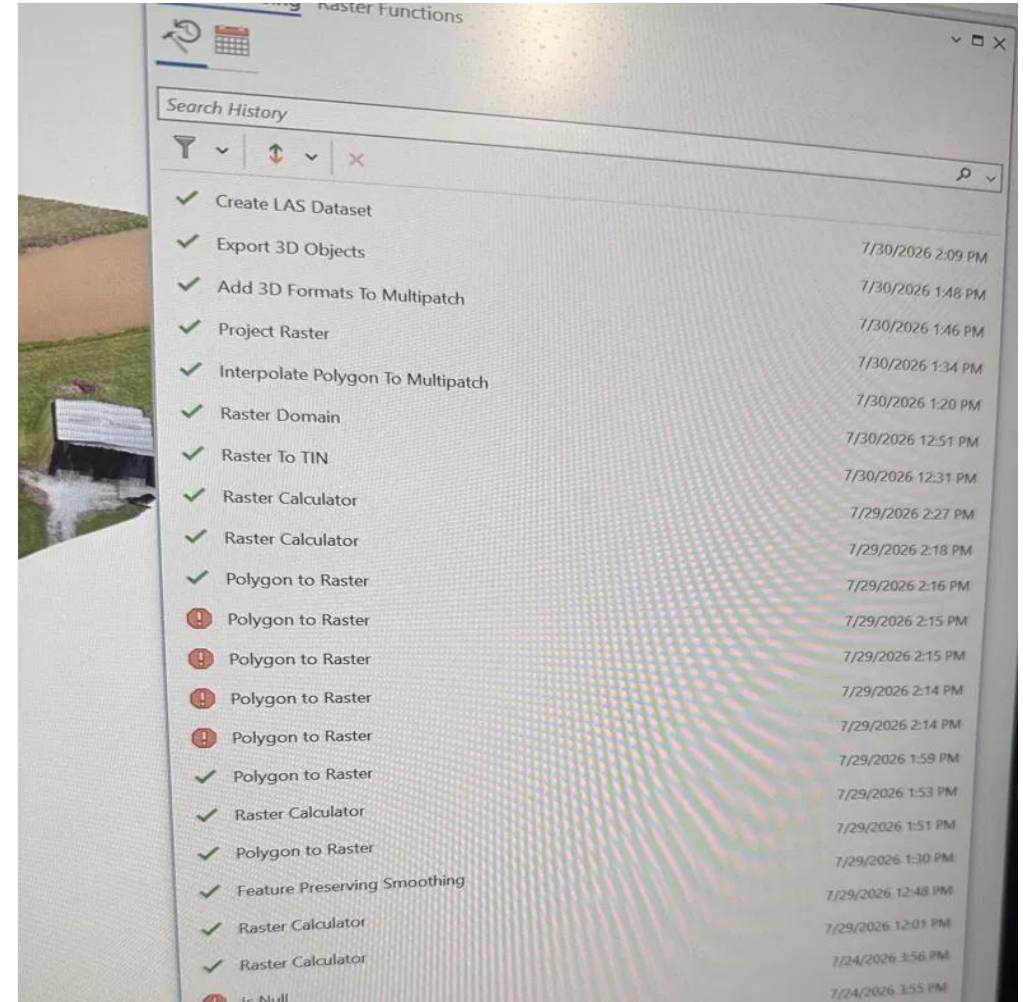


Geoprocessing Chain to 3D OBJ Export

Executed in ArcGIS Pro 3.6, 30 July 2026

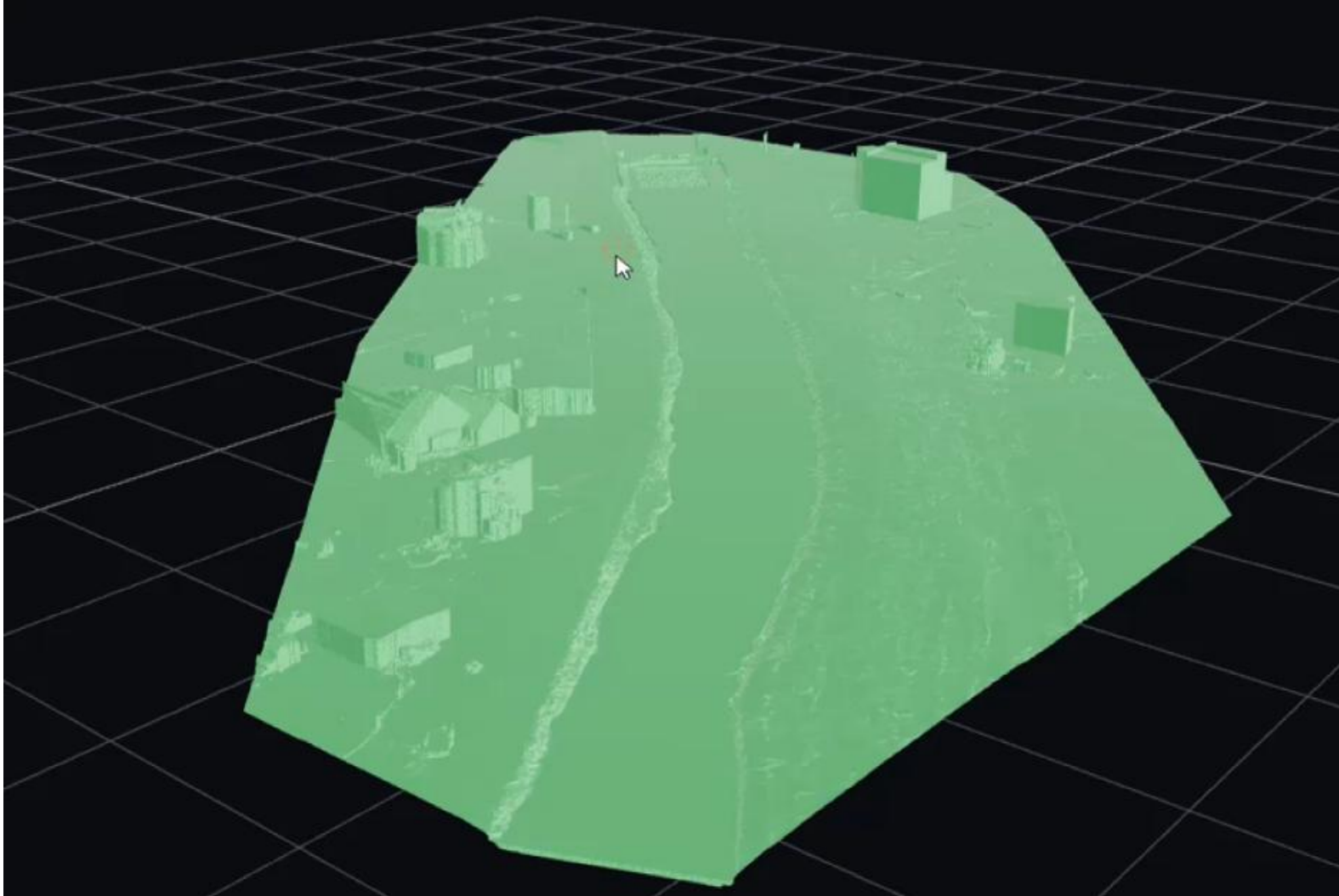
	Tool	Purpose
1	Raster Calculator	Finalize the hydro-flattened elevation surface
2	Raster To TIN	Convert the raster surface to a triangulated irregular network
3	Raster Domain	Generate the boundary polygon of the raster extent
4	Interpolate Polygon To Multipatch	Build multipatch geometry from the TIN within the domain
5	Project Raster	Align the coordinate system ahead of export
6	Add 3D Formats To Multipatch	Attach the OBJ format to the multipatch feature class
7	Export 3D Objects	Write the OBJ model files to disk

The sequence above is the working chain, recovered from the ArcGIS Pro geoprocessing history.



Export Result: 3D Surface Model in OBJ Format

Written from the multipatch feature class, ArcGIS Pro 3.6, 30 July 2026



Deliverables, Archive and Closing Remarks

Deliverables

- Digital surface model, raw and hydro-flattened, as GeoTIFF
- Orthomosaic as GeoTIFF
- Dense point cloud in LAS format
- ArcGIS Pro 3.6 project with full geoprocessing history
- 3D surface model in OBJ format
- Web scene on ArcGIS Online, supporting deliverable

Archive location

Lamar University OneDrive

OneDrive - Lamar University\Feilin Lai's files - 5_CAWAQ_LNVA\Munta

Known limitations

- The vertical coordinate system was not fully resolved. Elevation was reported in units inconsistent with the metric horizontal system, and the OBJ was exported in that state.
- No ground control points were surveyed, so absolute vertical accuracy is unverified.
- The model represents terrain only. Vegetation and structures were not captured as geometry.

Recommended next steps

- Define and apply the correct vertical coordinate system before the model is used for measurement
- Establish ground control at the site ahead of any repeat flight
- Retain the CAWAQ_LNVA project structure so repeat flights can be added as a time series